

SQL Subquery Practice Set

Use Case: Retail Sales Tracking System

1. Table Schema

```
CREATE TABLE sales (  
  sale_id          INT PRIMARY KEY,  
  product_category VARCHAR(50),  
  product_name     VARCHAR(50),  
  salesperson      VARCHAR(50),  
  region           VARCHAR(20),  
  sale_amount      DECIMAL(10,2),  
  quantity         INT,  
  sale_date        DATE  
);
```

2. Sample Data (24 Rows)

```
INSERT INTO sales VALUES  
(1, 'Electronics', 'Laptop', 'Alice', 'North', 75000, 2, '2024-01-05'),  
(2, 'Electronics', 'Phone', 'Bob', 'South', 25000, 5, '2024-01-10'),  
(3, 'Clothing', 'Jacket', 'Alice', 'East', 8000, 3, '2024-01-15'),  
(4, 'Furniture', 'Chair', 'Charlie', 'West', 12000, 4, '2024-01-20'),  
(5, 'Electronics', 'Tablet', 'Bob', 'North', 35000, 2, '2024-02-03'),  
(6, 'Clothing', 'Shirt', 'Charlie', 'South', 3000, 10, '2024-02-08'),  
(7, 'Furniture', 'Table', 'Alice', 'East', 22000, 1, '2024-02-12'),  
(8, 'Electronics', 'Laptop', 'Charlie', 'West', 80000, 1, '2024-02-18'),  
(9, 'Clothing', 'Jeans', 'Bob', 'North', 6000, 5, '2024-02-25'),  
(10, 'Furniture', 'Sofa', 'Alice', 'South', 45000, 2, '2024-03-02'),  
(11, 'Electronics', 'Phone', 'Charlie', 'East', 28000, 3, '2024-03-07'),  
(12, 'Clothing', 'Jacket', 'Bob', 'West', 9500, 2, '2024-03-14'),  
(13, 'Furniture', 'Wardrobe', 'Charlie', 'North', 32000, 1, '2024-03-19'),  
(14, 'Electronics', 'Headphones', 'Alice', 'South', 5000, 8, '2024-03-25'),  
(15, 'Clothing', 'Shirt', 'Alice', 'East', 3500, 12, '2024-04-01'),  
(16, 'Furniture', 'Bookshelf', 'Bob', 'West', 15000, 3, '2024-04-06'),  
(17, 'Electronics', 'Tablet', 'Charlie', 'North', 38000, 2, '2024-04-11'),  
(18, 'Clothing', 'Jeans', 'Alice', 'South', 7000, 4, '2024-04-16'),  
(19, 'Furniture', 'Chair', 'Bob', 'East', 13000, 5, '2024-04-21'),  
(20, 'Electronics', 'Laptop', 'Alice', 'West', 72000, 1, '2024-04-26'),  
(21, 'Clothing', 'Jacket', 'Charlie', 'North', 10000, 3, '2024-05-01'),  
(22, 'Furniture', 'Table', 'Bob', 'South', 24000, 2, '2024-05-06'),  
(23, 'Electronics', 'Phone', 'Alice', 'East', 27000, 4, '2024-05-11'),  
(24, 'Clothing', 'Shirt', 'Bob', 'West', 4000, 8, '2024-05-16');
```

Visual preview of all rows:

sale_id	product_category	product_name	salesperson	region	sale_amount	quantity	sale_date
1	Electronics	Laptop	Alice	North	75000	2	2024-01-05
2	Electronics	Phone	Bob	South	25000	5	2024-01-10
3	Clothing	Jacket	Alice	East	8000	3	2024-01-15
4	Furniture	Chair	Charlie	West	12000	4	2024-01-20

5	Electronics	Tablet	Bob	North	35000	2	2024-02-03
6	Clothing	Shirt	Charlie	South	3000	10	2024-02-08
7	Furniture	Table	Alice	East	22000	1	2024-02-12
8	Electronics	Laptop	Charlie	West	80000	1	2024-02-18
9	Clothing	Jeans	Bob	North	6000	5	2024-02-25
10	Furniture	Sofa	Alice	South	45000	2	2024-03-02
11	Electronics	Phone	Charlie	East	28000	3	2024-03-07
12	Clothing	Jacket	Bob	West	9500	2	2024-03-14
13	Furniture	Wardrobe	Charlie	North	32000	1	2024-03-19
14	Electronics	Headphones	Alice	South	5000	8	2024-03-25
15	Clothing	Shirt	Alice	East	3500	12	2024-04-01
16	Furniture	Bookshelf	Bob	West	15000	3	2024-04-06
17	Electronics	Tablet	Charlie	North	38000	2	2024-04-11
18	Clothing	Jeans	Alice	South	7000	4	2024-04-16
19	Furniture	Chair	Bob	East	13000	5	2024-04-21
20	Electronics	Laptop	Alice	West	72000	1	2024-04-26
21	Clothing	Jacket	Charlie	North	10000	3	2024-05-01
22	Furniture	Table	Bob	South	24000	2	2024-05-06
23	Electronics	Phone	Alice	East	27000	4	2024-05-11
24	Clothing	Shirt	Bob	West	4000	8	2024-05-16

Q1. Find all sales where the sale amount is higher than the overall average sale amount across all transactions. Display sale_id, product_category, salesperson, and sale_amount, sorted by sale_amount in descending order.

► Expected Output:

sale_id	product_category	salesperson	sale_amount
8	Electronics	Charlie	80000
1	Electronics	Alice	75000
20	Electronics	Alice	72000
10	Furniture	Alice	45000
17	Electronics	Charlie	38000
5	Electronics	Bob	35000
13	Furniture	Charlie	32000
11	Electronics	Charlie	28000
23	Electronics	Alice	27000
2	Electronics	Bob	25000

Q2. Retrieve the complete details of the transaction that recorded the single highest sale amount.

► Expected Output:

sale_id	product_category	product_name	salesperson	region	sale_amount	quantity	sale_date
8	Electronics	Laptop	Charlie	West	80000	1	2024-02-18

Q3. Find all sales made by the salesperson who has the highest total number of transactions. Display sale_id, salesperson, product_name, and sale_amount, sorted by sale_id.

► Expected Output:

sale_id	salesperson	product_name	sale_amount
1	Alice	Laptop	75000
3	Alice	Jacket	8000
7	Alice	Table	22000
10	Alice	Sofa	45000
14	Alice	Headphones	5000
15	Alice	Shirt	3500
18	Alice	Jeans	7000
20	Alice	Laptop	72000
23	Alice	Phone	27000

Q4. Find all sales of products that have been sold by more than 2 different salespersons. Display sale_id, product_name, salesperson, and sale_amount, sorted by sale_id.

► Expected Output:

sale_id	product_name	salesperson	sale_amount
2	Phone	Bob	25000
3	Jacket	Alice	8000
6	Shirt	Charlie	3000
11	Phone	Charlie	28000
12	Jacket	Bob	9500
15	Shirt	Alice	3500
21	Jacket	Charlie	10000
23	Phone	Alice	27000
24	Shirt	Bob	4000

Q5. Find all sales from regions whose average sale amount is above the overall average sale amount of all regions. Display sale_id, product_category, salesperson, region, and sale_amount, sorted by sale_id.

► Expected Output:

sale_id	product_category	salesperson	region	sale_amount
1	Electronics	Alice	North	75000
4	Furniture	Charlie	West	12000
5	Electronics	Bob	North	35000
8	Electronics	Charlie	West	80000
9	Clothing	Bob	North	6000
12	Clothing	Bob	West	9500
13	Furniture	Charlie	North	32000
16	Furniture	Bob	West	15000
17	Electronics	Charlie	North	38000
20	Electronics	Alice	West	72000
21	Clothing	Charlie	North	10000
24	Clothing	Bob	West	4000

Q6. Find all sales where the sale amount is higher than every individual sale amount in the 'Furniture' category. Display sale_id, product_name, product_category, and sale_amount, sorted by sale_id.

► Expected Output:

sale_id	product_name	product_category	sale_amount
1	Laptop	Electronics	75000
8	Laptop	Electronics	80000
20	Laptop	Electronics	72000

Q7. Find all sales where the individual sale amount is above the average sale amount of its own product category. Display sale_id, product_category, salesperson, and sale_amount, sorted by sale_id.

► Expected Output:

sale_id	product_category	salesperson	sale_amount
1	Electronics	Alice	75000
3	Clothing	Alice	8000
8	Electronics	Charlie	80000
10	Furniture	Alice	45000
12	Clothing	Bob	9500
13	Furniture	Charlie	32000
18	Clothing	Alice	7000
20	Electronics	Alice	72000
21	Clothing	Charlie	10000
22	Furniture	Bob	24000

Q8. For each salesperson, find the transaction where they recorded their personal highest sale amount. Display sale_id, salesperson, product_name, and sale_amount, sorted by sale_amount in descending order.

► Expected Output:

sale_id	salesperson	product_name	sale_amount
8	Charlie	Laptop	80000
1	Alice	Laptop	75000
5	Bob	Tablet	35000

Q9. Find all sales where the sale amount exceeds the average sale amount of the salesperson who made that sale. Display sale_id, salesperson, product_category, and sale_amount, sorted by salesperson and then by sale_id.

► Expected Output:

sale_id	salesperson	product_category	sale_amount
1	Alice	Electronics	75000
10	Alice	Furniture	45000
20	Alice	Electronics	72000
2	Bob	Electronics	25000
5	Bob	Electronics	35000
22	Bob	Furniture	24000
8	Charlie	Electronics	80000
13	Charlie	Furniture	32000
17	Charlie	Electronics	38000

Q10. Find all sales where the quantity sold is greater than the average quantity sold in the same region. Display sale_id, product_category, salesperson, region, and quantity, sorted by sale_id.

► Expected Output:

sale_id	product_category	salesperson	region	quantity
4	Furniture	Charlie	West	4
6	Clothing	Charlie	South	10
9	Clothing	Bob	North	5
14	Electronics	Alice	South	8
15	Clothing	Alice	East	12
19	Furniture	Bob	East	5
21	Clothing	Charlie	North	3
24	Clothing	Bob	West	8